

1E-E

## EEL1010 Introduction to Electrical Engineering – Minor Exam

Duration: 2 hrs

Total Marks: 40

Date: 22/02/2025

All questions are compulsory.

Q0. What is your batch no.? Write it at the top of the answer sheet.

Q1(a) Draw the input and output characteristics of the configuration (Fig. 1) with proper labeling. 4

Q1(b) Show how you will calculate the desirable  $V_{CQ}$  and  $I_{CQ}$  on the load line (Fig. 1) 5

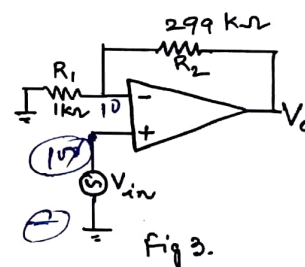
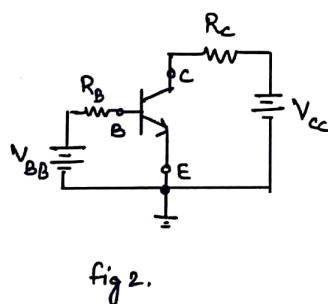
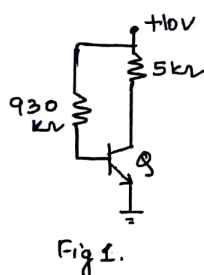
Q2 For the circuit in Fig. 2, describe and draw how  $I_C$  and  $V_{CE}$  change when Qpoint is tuned in different regions → Active, Saturation and Cut off 6

Q3 The op amp in Fig. 3 has open loop gain of 1000. If  $V_{in} = 10mV$ , then what is the output voltage? Will it be inverted or same? 5+1

Q4 Explain the functioning and PIV of a centre tap transformer, justifying PIV value. 5

Q5 Design a summing amplifier so the input voltages 100mV, 200mV and 300mV all get summed up. Assume the values of resistances in the circuit are same and Derive and prove that your circuit behaves like a summing amplifier. 6

Q6 Write the truth table and complete the gate implementation for the function  $F = A'C + BC'$ . Show its NAND equivalent circuit. 2+2+4



$$\frac{10 - 0}{R_1} = \frac{10 - V_o}{299}$$

$$\frac{299 \times 10}{R_1} = \frac{V_o - 10}{V_o - 10}$$

# EEL1010 Introduction to Electrical Engineering

## Major exam

Date: 27.04.2025, 9 am  
Max Marks: 75

Total Time: 3 hours

All questions are compulsory and subjective. Draw diagrams or write equations wherever necessary.

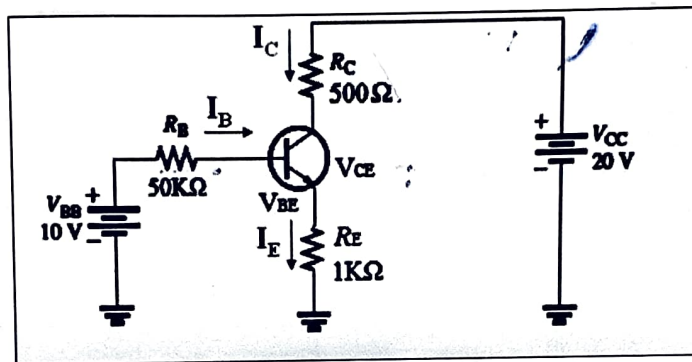
What is your Batch no.? Please write it at the top of your answer sheet.

### Part A

Q1. Determine the Q point for the circuit shown below and answer the questions that follow. The current gain of the transistor,  $\beta = 150$  and  $V_{BE} = 0.7V$ .

- Name the configuration of the transistor (CB, CC, CE)?
- Is it npn or pnp?
- Draw the DC load line

(1+1+4)=6

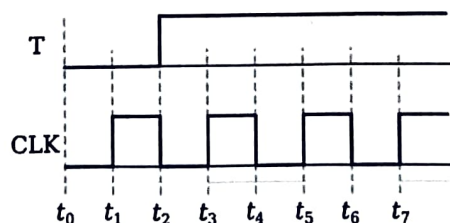


Q2. (a) Minimize the following function in SOP minimal form using K maps  $F(A,B,C,D) = \Sigma(1,2,6,7,8,13,14,15) + d(0,3,5,11)$

- Draw the logical circuits with all necessary gates
- Draw the logical circuits with only NAND gates
- Deduce the POS minimal form of the same function using K-maps

(3+2+2+2) = 9

Q3. Draw the timing diagram of Q and Q' for the given diagram of T flip flop. Assume positive edge-triggered clock. (2)



$$I_C = \beta I_B$$

$$\frac{I_E}{\beta} = I_B$$

1

$$F = B\bar{C}\bar{D} + \bar{A}\bar{B} + BD + CB$$

$$(A' + B')$$

$$A \cdot B$$

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## Major exam

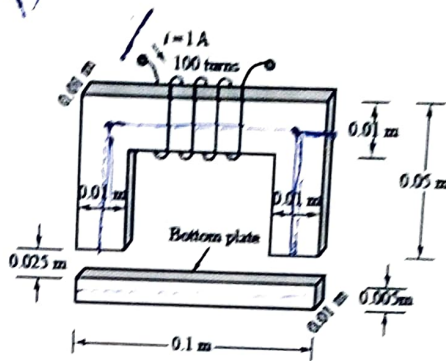
Date: 27.04.2025, 9 am  
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Q4. Explain how SR latch is made with NOR gates and what is done to avoid the indeterminate state. Support your answers with logic circuits and characteristic table. (5+8)=13

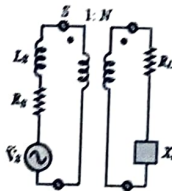
### Part B

- Q5. A) Define flux linkage. Derive expression for self-inductance and mutual inductance for a magnetic circuit of your choice.  
B) Describe the nonlinear properties of magnetic material. Mention the associated loss in the magnetic circuit.  
C) Compute the equivalent reluctance of the magnetic circuit of figure below and the flux density established in the bottom bar of the structure. [ $\mu_r = 10000$ ,  $N = 100$  turns,  $i = 1$  A]



(2+6) + (6+2) + 6=22

- Q6. A) Derive an expression for relation between currents in primary and secondary side for ideal transformer.  
B) Derive an equivalent circuit of practical transformer with all components referred to secondary side with proper justification of each component.  
C) Find the transformer turns ratio and load reactance that results in maximum power transfer in the circuit below: [ $\vec{V}_s = 240 \angle 0^\circ$  V,  $R_s = 10 \Omega$ ,  $L_s = 0.1$  H,  $R_L = 400 \Omega$ ,  $\omega = 377$  rad/s]



$L = N_1 N_2$

$\frac{A}{1} = L$

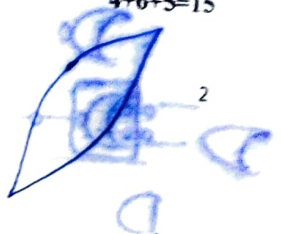
4+6+5=15

$P = (N_1 I_1 + N_2 I_2)$

$\frac{P}{P} = \phi \frac{\mu_0 \mu_r N_1^2}{l}$

$\mu_0 \mu_r \left( \frac{N_1}{l} \right)^2$

$N_2 (\phi)$



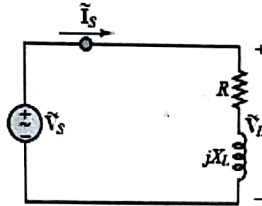
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## Major exam

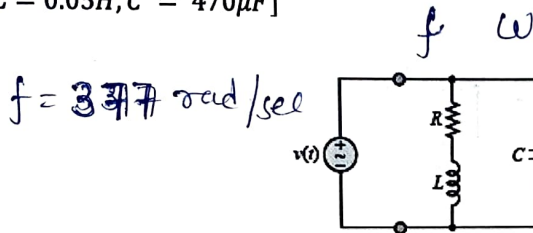
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Max Marks: 75

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7. A) Calculate the complex power for the below circuit and correct the power factor to unity by connecting a parallel reactance to the load.  $[\vec{V}_s = 117\angle 0^\circ \text{ V}, R_L = 50\Omega, X_L = 86.7\Omega]$



B) Compute the average power dissipated by the load of figure below:  $[\vec{V}_s = 110\angle 0^\circ \text{ V}, R = 10\Omega, L = 0.05\text{H}, C = 470\mu\text{F}]$



$f = 377 \text{ rad/sec}$

$$\frac{(R + jX)(8 + j4)}{(S + R) + j}$$

$4 + 4 = 8$

$\frac{V^2}{2} \frac{(25)^2}{(10)^2} = \frac{(110)^2}{2} \frac{25}{100} = 1501.25 \text{ W}$

\*\*\*\*\*END OF QUESTION PAPER\*\*\*\*\*

$$\frac{(R + jX)(5 + jV)}{(R + 5) + (X + V)j}$$

$$\frac{(R + jX)(-jY)}{(R + jX - jY)}$$

$R \times 50 + X \times -$

$$25 - X V + (R + X 5)j$$